FILE SYSTEM AND SECONDARY STORAGE STRUCTURES

MODULE 5:

FileSystem: FileConcept,AccessMethods,Directory and DiskStructure, File SystemMounting,FileSharing.

SecondaryStorageStructures: MassStorageStructure,DiskStructure,Disk Attachment,DiskScheduling.

File Concept

A file is a collection of similar records.

The file is treated as a single entity by users and applications and may be referred by name.

Files have unique file names and may be created and deleted.

Restrictions on access control usually apply at the file level. A file is a container for a collection of information.

The filemanager provides a protection mechanism to allow users administrator how processes executing on behalf of different users can access the information in a file.

File protection is a fundamental property of files because it allows different people to store their information on a shared computer. File represents programs and data.

Data files may be numeric, alphabetic, binary or alpha numeric. Files may be free form, such as text files.

In general, file is sequence of bits, bytes, lines or records.

A common technique for implementing file types is to include the type as part of the file name. the name is split into two parts a name and an extension. Various file types are shown in the following table.

File Type	Usual extension	Function
executable	exe, com, bin	read to run machine language program
object	obj, O	compiled, machine language, not linked
source Code	c, cc, java	source code in various languages
batch	bat, sh	commands to the command interpreter
text	txt, doc	textual data, documents
word processor	wp, tex, rtf, doc	various word-processor formats
library	lib, dll, mpeg, mov	libraries of routines for programmers
print or view	arc, zip, tar	ASCII or binary file in a format for printing or viewing
archieve	arc, zip, tar	related files grouped into one file, sometimes compressed for archiving or storage
multimedia	mpeg, mov, rm	binary file containing audio or A/V information

File Structure

- A file is a logical unit of information.
- They are produced by processes.
- The operating system manages files.
- While creating a file, a name is assigned to it.
- After this process has terminated, the file exists and remains accessible to other processes.
- The name of a file has two parts, separated by a dot.
- Example:

  code.cpp is a c++ program file with name code and extension cpp.
- The name of the file before the dot is a label for the file's identification, while the part after the dot is called the file extension which indicates the type of the file

Attributes of a File:

Name : It is the only information which is in human-readable form.

Identifier : The file is identified by a unique tag(number) within file system.

Type : It is needed for systems that support different types of files.

Location: Pointer to file location on device.

Size: The current size of the file.

Protection: This controls and assigns the power of reading, writing, executing.

Time, date, and user identification: This is the data for protection, security, and usage monitoring.

File Operations

File Operations

File is an **abstract data type** and the OS provides a number of minimal operations on it.

- **Create**
  - *Allocate space and then make an entry in the directory*
- **Write**
  - *Requires name of the file, and the information to be written*
  - *Search the directory to find file's location.*
  - *Keep a write-pointer to the location in the file*
  - *Update the pointer after each write*
- **Read**
  - *Requires name of the file, and the information to be read*
  - *Search the directory to find file's location.*
  - *Keep a read-pointer to the location in the file*
  - *Update the pointer after each read*
- **Reposition within file**
  - *Change the value of the file-position pointer*
- **Delete**
  - *Deallocate the space and remove the entry*
- **Truncate**
  - *Change the allocated space to zero, and deallocate its space*

File-position pointer

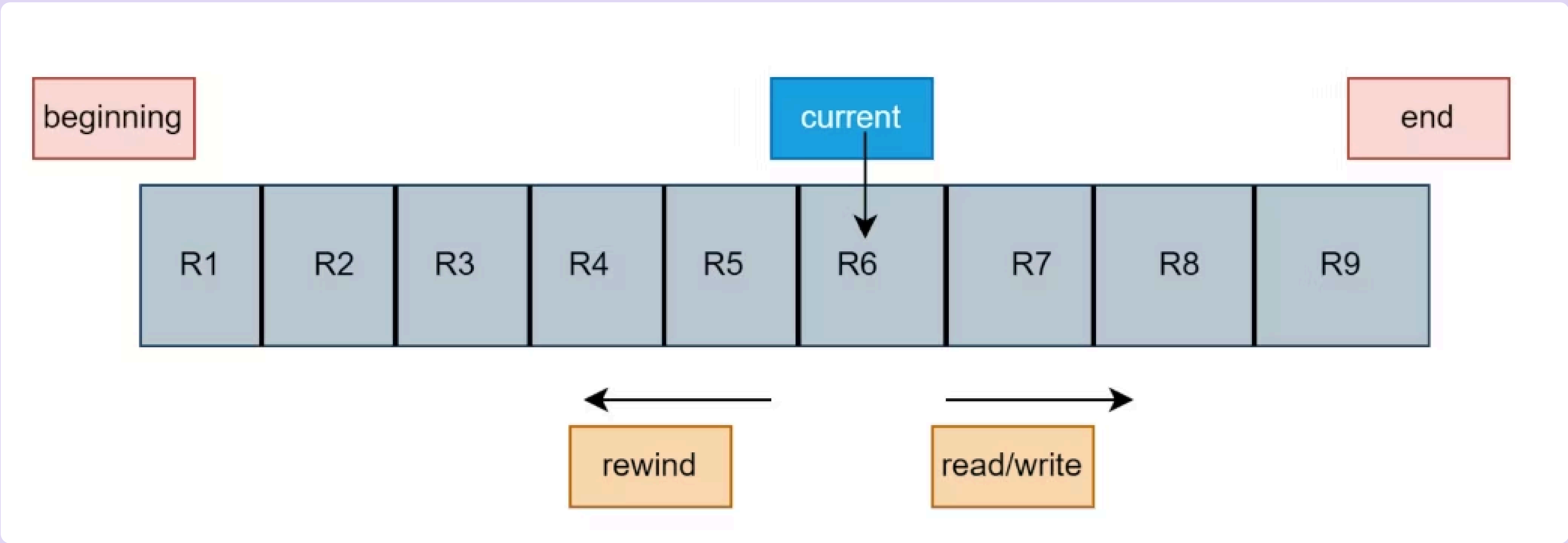
File Access Method

Files store information. When it is used, this information must be accessed and read into computer memory. The information in the file can be accessed in several ways.

Sequential Access Method:

It is the simplest access method. Information in the file is processed in order i.e. one record after another. A process can read all the data in a file in order starting from beginning but can't skip & read arbitrarily from any location. Sequential files can be rewound. It is convenient when storage medium was magnetic tape rather than disk.

Example:



The figure represents a file. The current pointer is pointing to the record currently being accessed. In the sequential access method, the current pointer cannot directly jump to any record. It has to "cross" every record that comes in its path. Suppose there are nine records in the file from R1 to R9. The current pointer is at record R6. If we want to access record R8, we have to first access record R6 and record R7. This is one of the major disadvantages of the sequential access method.

The sequential access method has three operations:

- **Read next:** It will read the next record in the file. The file pointer (current pointer) will advance to the next record. It is similar to how we traverse the nodes in a linked list.
- **Write next:** This operation is used when some more information is to be included in the file. A new node (a record) will be added at the end of the file. The end pointer will now point to the node that has been added, marking it as the new end of the file. It is similar to adding a new node at the end of a linked list.
- **Rewind:** This will bring the read and write pointers to the beginning of the file.

Direct Access Method

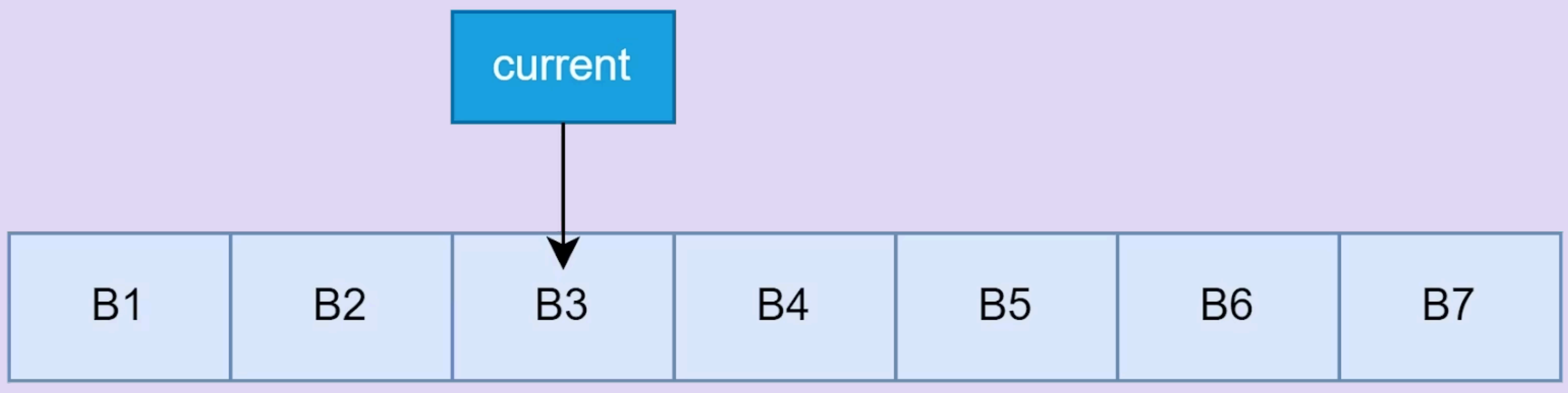
Sometimes it is not necessary to process every record in a file.

· It is not necessary to process all the records in the order in which they are present in the memory. In all such cases, direct access is used.

· The disk is a direct access device which gives us the reliability to random access of any file block.

· In the file, there is a collection of physical blocks and the records of that blocks.

· Example: Databases are often of this type since they allow query processing that involves immediate access to large amounts of information. All reservation systems fall into this category.

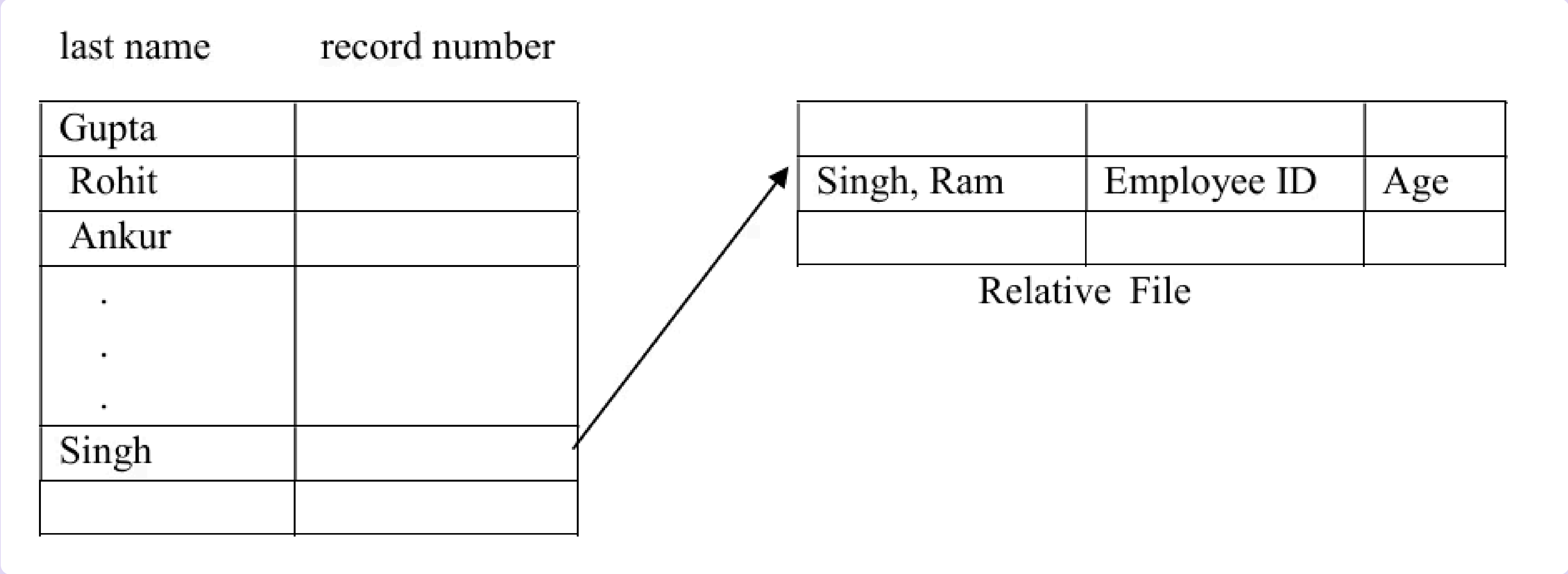


The direct access method has the following operations:

- **Read n:** This operation is used to read the nth block. Read 6 would allow us to read block B6.
- **Write n:** This operation is used to write in the nth block.
- **Goto n:** This operation is used to directly access the nth block.

Indexed Access Method

In this method an index is created which contains a key field and pointers to the various blocks. To find an entry in the file for a key value, we first search the index and then use the pointer to directly access a file and find the desired entry. With large files, the index file itself may become too large to be keep in memory. One solution is to create an index for the index file. The primary index file would contain pointers to secondary index files, which would point to the actual data items. Figure shows a situation as implemented by VMS (Virtual Memory Storage) index and relative files.



Directory and Disk Structure

- On a computer, a directory is used to store, arrange, and segregate files and folders.
- It is similar to a telephone directory in that it just contains lists of names, phone numbers, and addresses rather than the real papers.
- It uses a hierarchical structure to organise files and directories.
- On many computers, directories are referred to as drawers or folders, much like a workbench or a standard filing cabinet in an office.

There are several logical structures of a directory, these are given below.

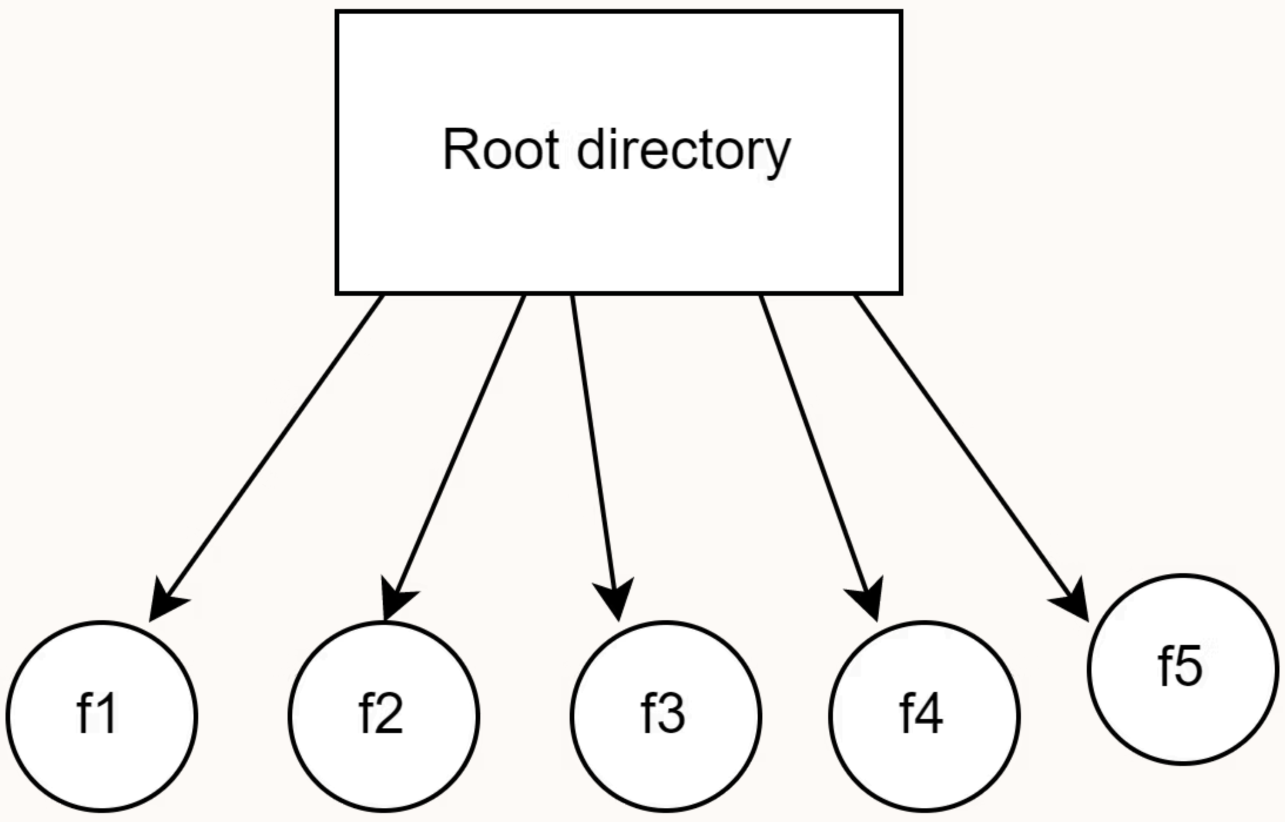
1. Single level directory
2. Two-level directory
3. Tree structure or hierarchical directory
4. Acyclic graph directory
5. General graph directory structure and Data Structure

Single-Level Directory Structure

It is the simplest directory structure. In a single-level directory, there is only one directory in the system, meaning there is only one folder, and all the files are stored in that single directory. There is no way to segregate important files from non-important files.

Implementation of a single-level directory is the simplest. However, there are various disadvantages of it.

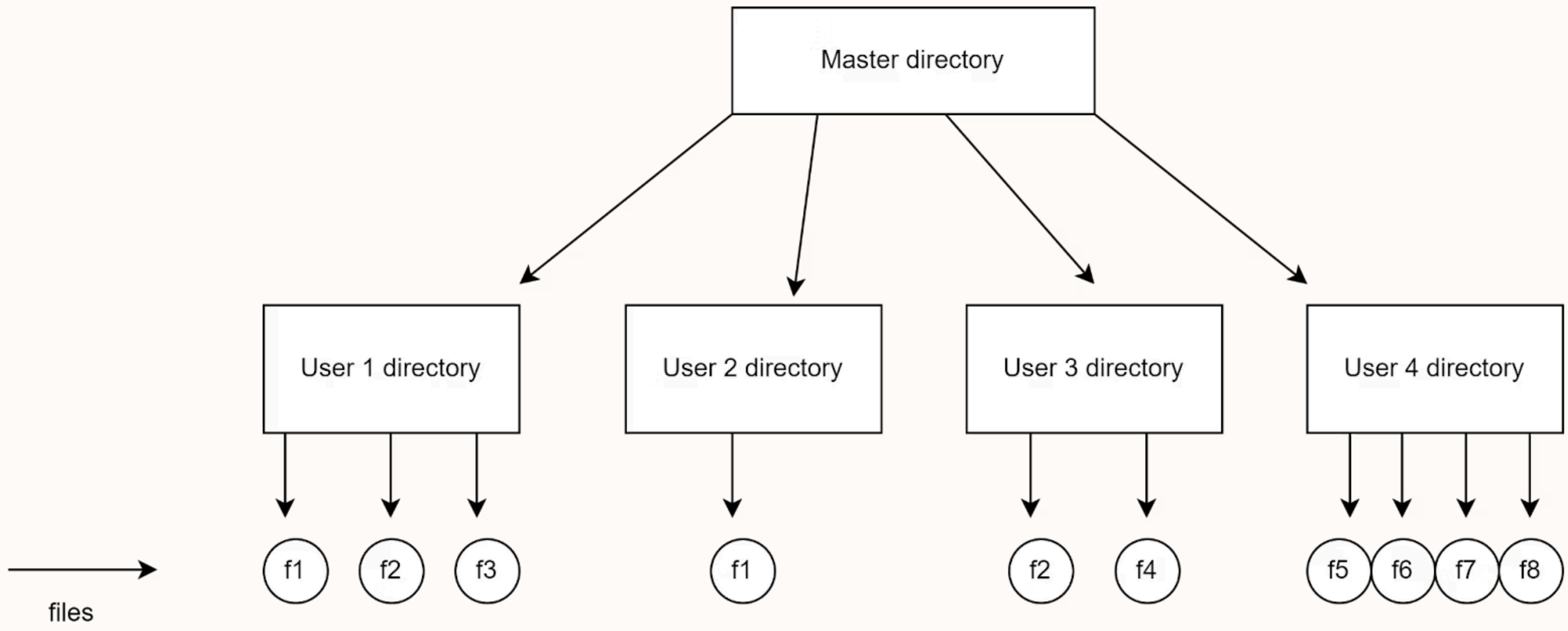
The pictorial representation of a single-level directory is given below. There is only one directory (root directory), and all the files are stored in the same directory. Here f1, f2, f3, f4, f5 represent the five different files. Practically it can be thought of as a structure where all the files are stored in the same folder.



Two-Level Directory Structure

In a two-level directory structure, there is a master node that has a separate directory for each user. Each user can store the files in that directory. It can be practically thought of as a folder that contains many folders, each for a particular user, and now each user can store files in the allocated directory just like a single level directory.

The pictorial representation of a two-level directory is shown below. For every user, there is a separate directory. At the next level, every directory stores the files just like a single-level directory. Although not very efficient, the two-level directory is better than a single-level directory structure.



Advantages of two-level directory

- Searching is very easy.
- There can be two files with the same name in two different user directories. Since they are not in the same directory, the same name can be used.
- Grouping is easier.
- A user cannot enter another user's directory without permission.
- Implementation is easy.

Disadvantages of two-level directory

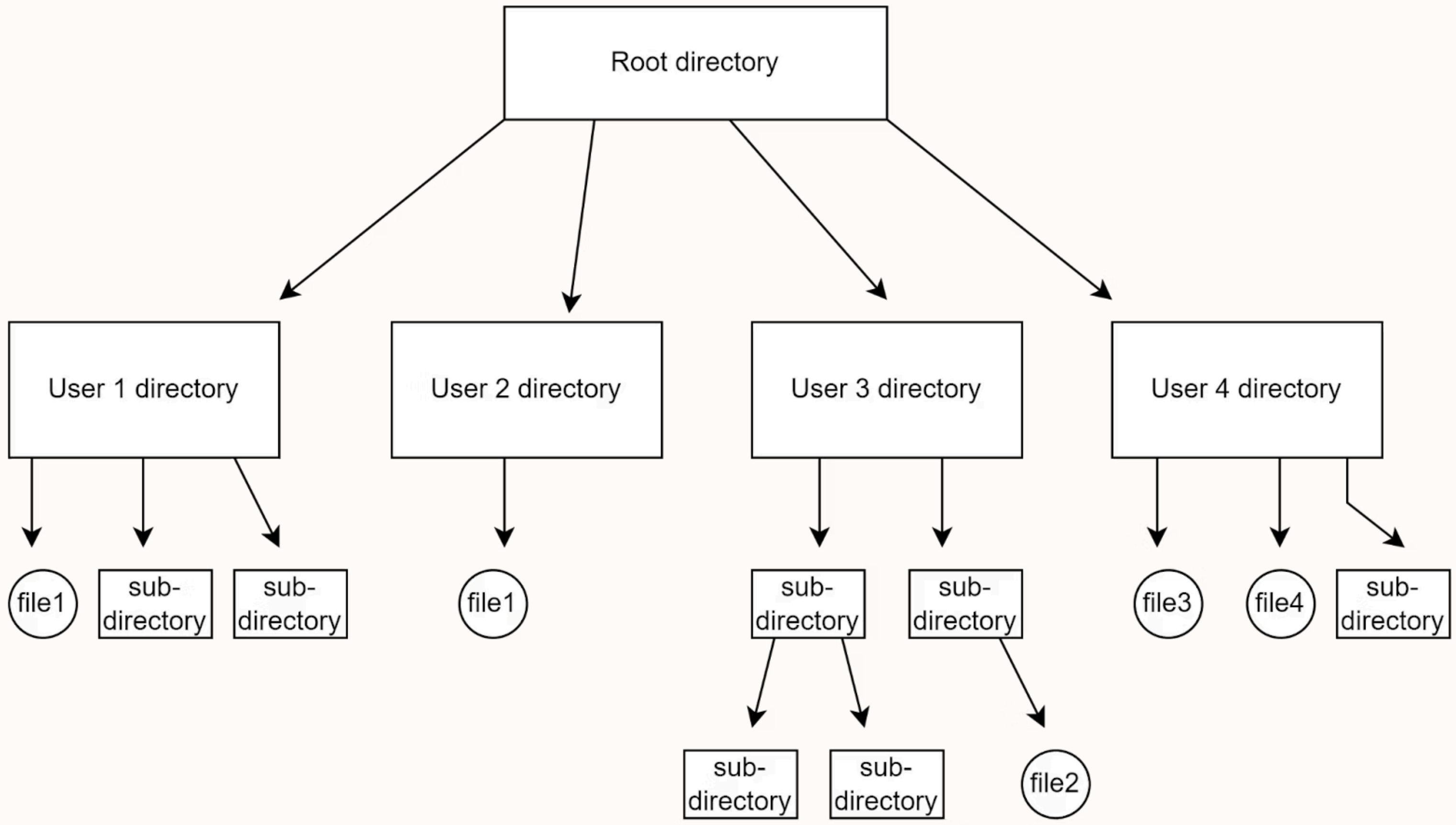
- One user cannot share a file with another user.
- Even though it allows multiple users, still a user cannot keep two same type files in a user directory.
- It does not allow users to create subdirectories.

Tree-Structured Directory Structure

This type of directory is used in our PCs. The biggest disadvantage of a two-level directory was that one could not create sub-directories in a directory. The tree-structured directory solved this problem. In a tree-structured directory, there is a root directory at the peak. The root directory contains directories for each user. The users can, however, create subdirectories inside their directory and also store the files.

This is how things work on our PCs. We can store some files inside a folder and also create multiple folders inside a folder.

The pictorial representation of a tree-structured directory is shown below. The root directory is highly secured, and only the system administrator can access it. We can see how there can be subdirectories inside the user directories. A user cannot modify the root directory data. Also, a user cannot access another user's directory.



Advantages of tree-structured directory

- Highly scalable compared to the previous two types of directories.
- Allows subdirectories inside a directory.
- Searching is easy.
- Allows grouping.
- Segregation of important and unimportant files is easy.

Disadvantages of tree-structured directory

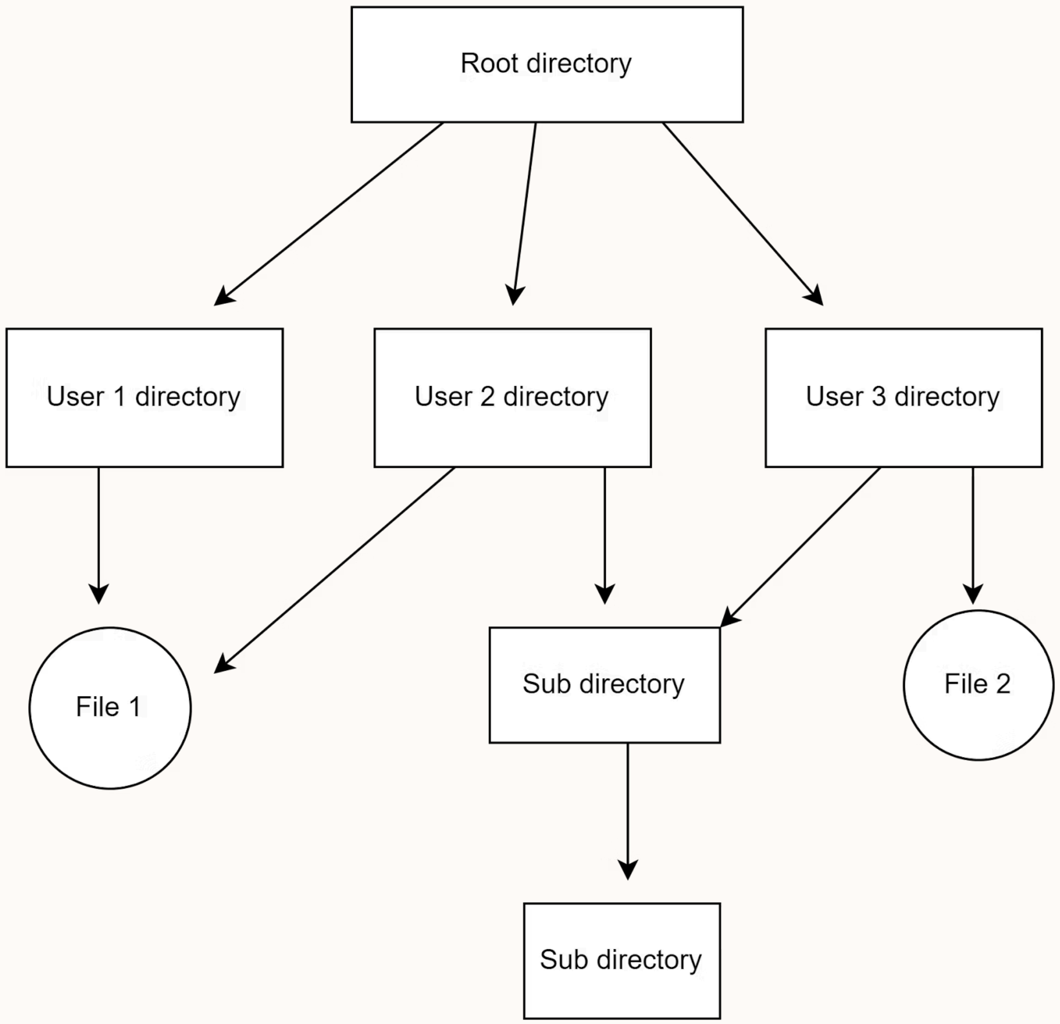
- As one user cannot enter another user's directory, this restricts sharing of files.
- Too many subdirectories may make the search complicated.
- Users cannot modify the root directory's data.
- Files might have to be saved in multiple directories in case all of them do not fit into one.

Acyclic Graph Directory Structure

In this type of directory, we can access a file or a subdirectory from multiple directories. Hence files can be shared between directories. It is designed in such a way that multiple directories point to a particular directory or file with the help of links.

A practical example of this is a doc file shared between two users. If any of the users makes a change in the file, the change is reflected for both the users.

Below is the pictorial representation of the acyclic-graph directory.



Advantages of acyclic- graph directory

- Allows sharing of files or subdirectories from more than one directory.
- Searching is very easy.
- Provides more flexibility to the users.

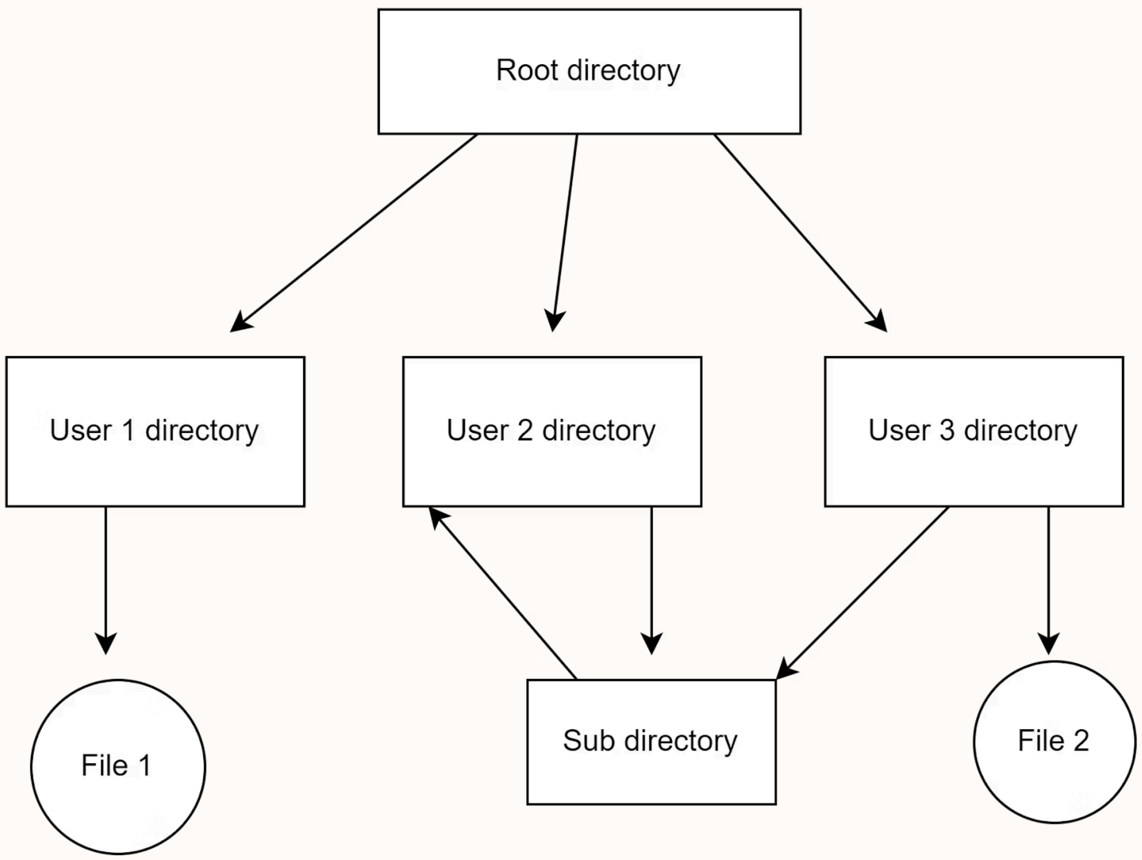
Disadvantages of acyclic-graph directory

- Harder to implement in comparison to the previous three.
- Since the files are accessed from multiple directories, deleting a file may cause some errors if the user is not cautious.
- If the files are linked by a hard link, then it is necessary to delete all the references to that file to permanently delete the file.

General-Graph Directory Structure

This is an extension to the acyclic-graph directory. In the general-graph directory, there can be a cycle inside a directory.

In the above image, we can see that a cycle is formed in the user 2 directory. Although it provides greater flexibility, it is complex to implement this structure.



Advantages of General-graph directory

- Compared to the others, the General-Graph directory structure is more flexible.
- Cycles are allowed in the directory for general-graphs.

Disadvantages of General-graph directory

- It costs more than alternative solutions.
- Garbage collection is an essential step here.

File System Mounting

Mounting is a process in which the operating system adds the directories and files from a storage device to the user's computer file system. The file system is attached to an empty directory, by adding so the system user can access the data that is available inside the storage device through the system file manager. Storage systems can be internal hard disks, external hard disks, USB flash drivers, SSD cards, memory cards, network-attached storage devices, CDs and DVDs, remote file systems, or anything else.

Terminologies used in File System Mounting

- **File System:** It is the method used by the operating system to manage data storage in a storage device. So, a user can access and organize the directories and files in an efficient manner.
- **Device name:** It is a name/identifier given to a storage partition. In windows, for example, "D:" in windows.
- **Mount point**: It is an empty directory in which we are adding the file system during the process of mounting.

Windows OS

In windows mounting is very easy for a user. When we connect the external storage devices, windows automatically detect the file system and mount it to the drive letter. Drive letter may be **D:** or **E:**.

Steps:

- Connect an external storage device to your PC.
- Windows detects the file system on the drive (e.g., FAT32 or NTFS) and assigns it a drive letter, such as "E:".
- You can access the derive by going through, THIS PC --> FILE EXPLORER -->"E:" drive
- Access the data.

File Sharing

File sharing is a productivity tool that enables a limited group of users to exchange data remotely. These can be shared with a single person, two people or even a whole company. File sharing allows users to share practically any type of file format including images, PowerPoint slides, text documents, audio and video files, and so on. When it comes to promoting cooperation and communication between people and organizations, file sharing is needed. It makes remote work possible and lessens the need for in-person meetings by letting users transfer files quickly and effortlessly between locations.

Working of File Sharing

- An administrator creates a folder and allows access to the designated users to arrange files in a corporate file-sharing solution.
- Usually, this entails making a group or groups and then adding the groups to the access control list of the folder.
- This enables the administrator to configure read/write access for specific users or entire groups as needed.
- Users can then access files by going to the directories that they can access. Files are hosted on a physical server or in the cloud, or they are kept on another computer.
- A user may often read and edit a saved file, save any changes, and have those changes show up when they access the file again.
- The most recent changes to a shared folder or file are visible to all users who have access to it.

Types of File Sharing

Below are four types of File Sharing

- **Cloud-Based File Sharing**: Cloud-Based File Sharing calls for storing files on an inaccessible server that is accessible from any internet-connected device. These services like **Dropbox, Google Drive,** and OneDrive allow users to upload and download data.
- **Peer-to-peer (P2P): Peer-to-peer** file sharing is the transmission of Files Users may exchange files with one another without a centralized server. These are equipotent, equally privileged members of the network that comprise a peer-to-peer network of nodes.
- **Sharing of Removable Media Files:** This type of file sharing requires the usage of physical storage devices, such as external hard drives or USB drives. By physically transferring the device to others, users may copy files onto it and share them with others.
- **Direct File Transfer:** Transferring data between two devices directly using a direct connection, such as Bluetooth or **Wi-Fi** Direct, is known as direct file transfer. File sharing between laptops and mobile devices often occurs via direct file transfer.

Secondary Storage Structures

Mass Storage Structure

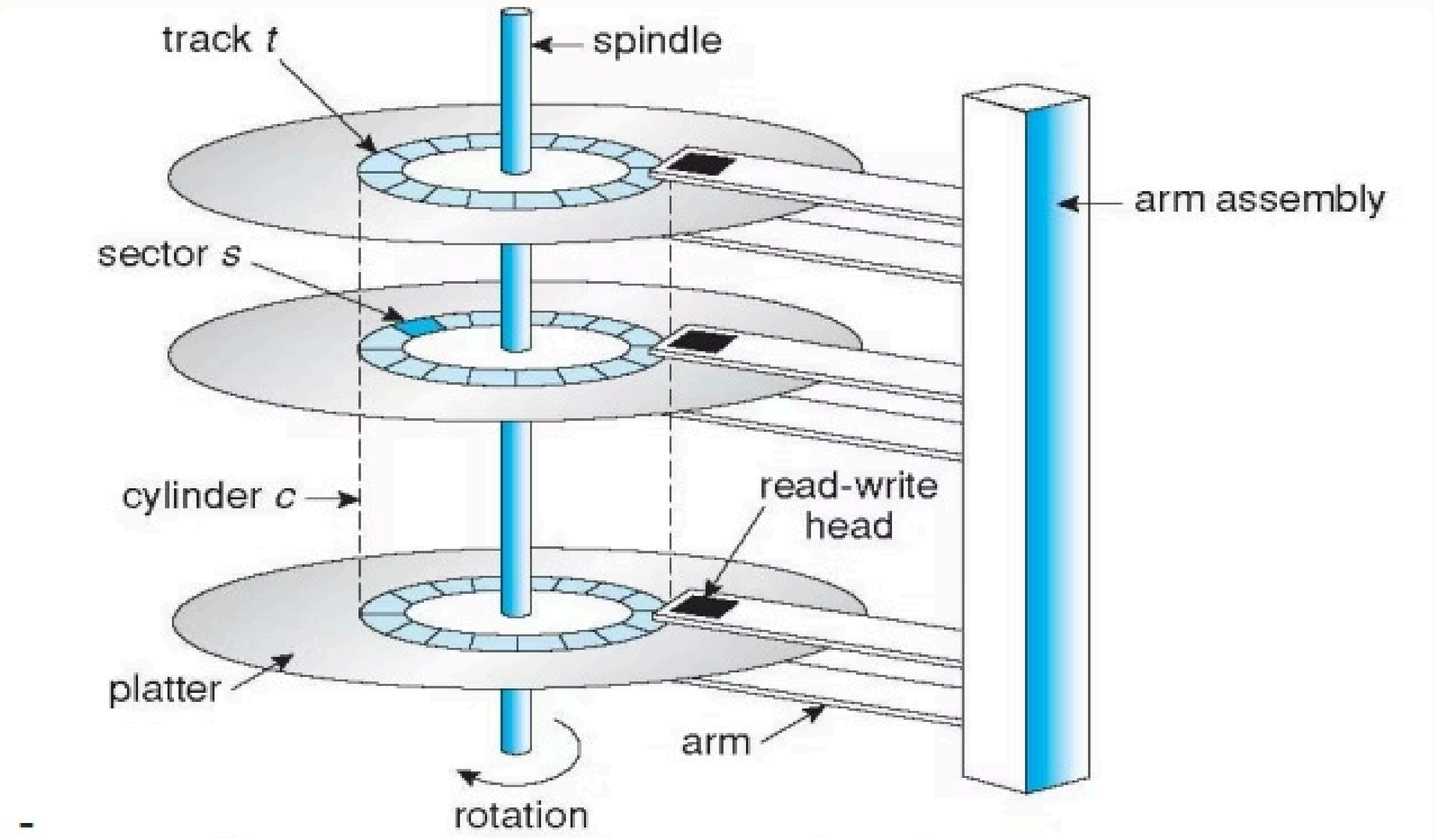


Figure 7.1 Moving-head disk mechanism.

- Magnetic disks provide the bulk of secondary storage for modern computer systems.
- Each disk platter has a flat circular shape, like a CD. Common platter diameters range from 1.8 to 5.25 inches.
- The two surfaces of a platter are covered with a magnetic material. The information stored by recording it magnetically on the platters.
- The surface of a platter is logically divided into circular tracks, which are subdivided into sectors.
- Sector is the basic unit of storage. The set of tracks that are at one arm position makes up a cylinder.
- The number of cylinders in the disk drive equals the number of tracks in each platter.
- There may be thousands of concentric cylinders in a disk drive, and each track may contain hundreds of sectors.

Some important terms:

- **Seek Time**:-Seek time is the time required to move the disk arm to the required track.
- **Rotational Latency (Rotational Delay)**:- Rotational latency is the time taken for the disk to rotate so that the required sector comes under the r/w head.
- **Positioning time** or random access time is the summation of seek time and rotational delay.
- **Disk Bandwidth**:- Disk bandwidth is the total number of bytes transferred divided by total time between the first request for service and the completion of last transfer.
- **Transfer rate** is the rate at which data flow between the drive and the computer. As the disk head flies on an extremely thin cushion of air, the head will make contact with the disk surface. Although the disk platters are coated with a thin protective layer, sometimes the head will damage the magnetic surface. This accident is called a head crash.

Magnetic Tapes

- Magnetic tape is a secondary-storage medium. It is a permanent memory and can hold large quantities of data.
- The time taken to access data (access time) is large compared with that of magnetic disk, because here data is accessed sequentially.
- When the nth data has to be read, the tape starts moving from first and reaches the nth position and then data is read from nth position. It is not possible to directly move to the nth position. So tapes are used mainly for backup, for storage of infrequently used information.
- A tape is kept in a spool and is wound or rewound past a read-write head. Moving to the correct spot on a tape can take minutes, but once positioned, tape drives can write data at speeds comparable to disk drives

Disk Structure

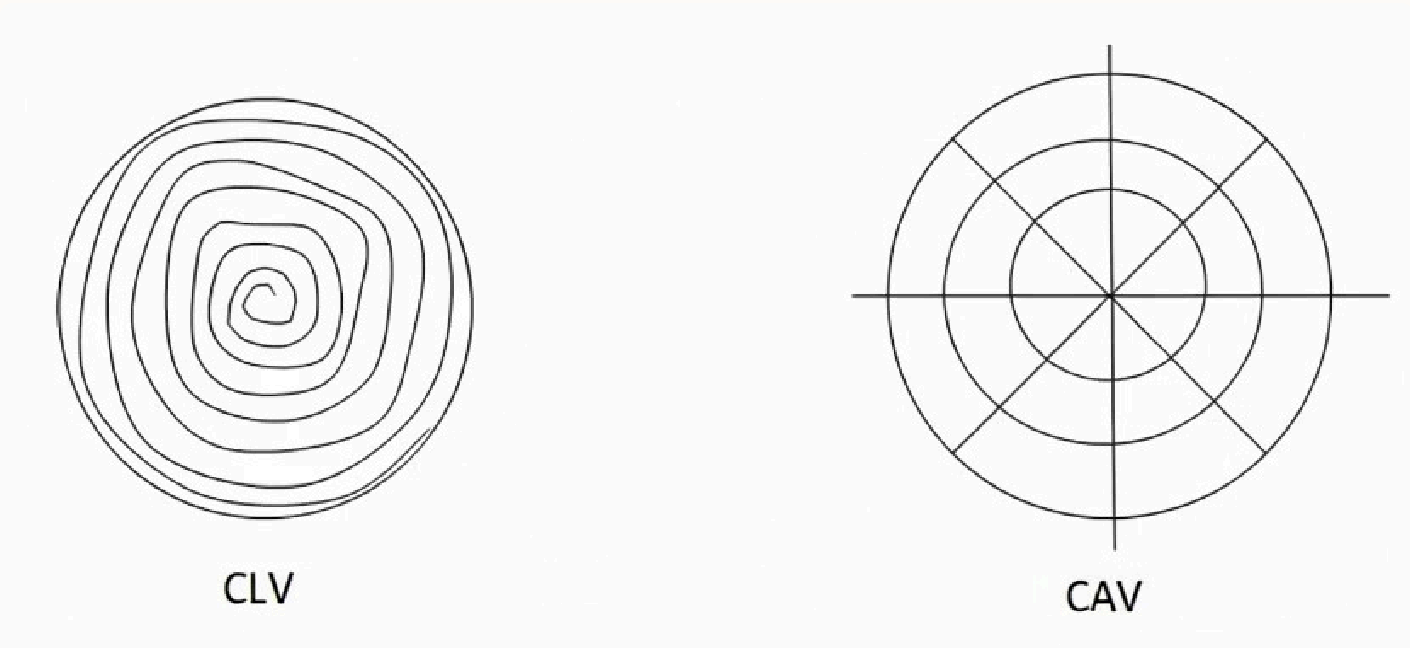
• Modern disk drives are addressed as a large one-dimensional array. The one- dimensional array of logical blocks is mapped onto the sectors of the disk sequentially.

• Sector 0 is the first sector of the first track on the outermost cylinder. The mapping proceeds in order through that track, then through the rest of the tracks in that cylinder, and then through the rest of the cylinders from outermost to innermost. The disk structure (architecture) can be of two types –

1. Constant Linear Velocity (CLV)
2. Constant Angular Velocity (CAV)

CLV – The density of bits per track is uniform. The farther a track is from the center of the disk, the greater its length, so the more sectors it can hold. As we move from outer zones to inner zones, the number of sectors per track decreases. This architecture is used in CD-ROM and DVD-ROM.

CAV – There is same number of sectors in each track. The sectors are densely packed in the inner tracks. The density of bits decreases from inner tracks to outer tracks to keep the data rate constant.



Disk Attachment

The disk attachment is a special feature that allows you to attach a disk (such as a USB flash drive) to your computer's hard drive. Users can use this feature to store files in the cloud or transfer files between computers on their network.

1

Host-Attached Storage

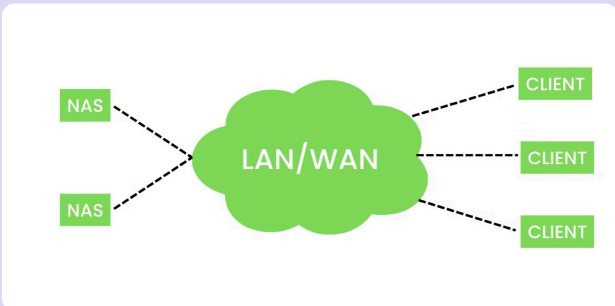
Host-attached storage (HAS) is a form of internal computer storage that can be attached to a host computer, such as a PC or server. Host-attached devices are often used for backup purposes and can include tape drives, optical drives, **hard disk drives** (HDDs), **solid state drives** (SSDs), **USB flash drives,** and other similar media.

A common example of host-attached storage is the use of an external USB flash drive for data transfer between computers.

2

Network-Attached Storage (NAS)

A Network-Attached Storage (NAS) is a computer that is connected to a network and shared by multiple users. NAS can also be called a file server, or storage appliance. The term "storage" refers to any device that stores data; this includes hard drives and flashes memory modules.

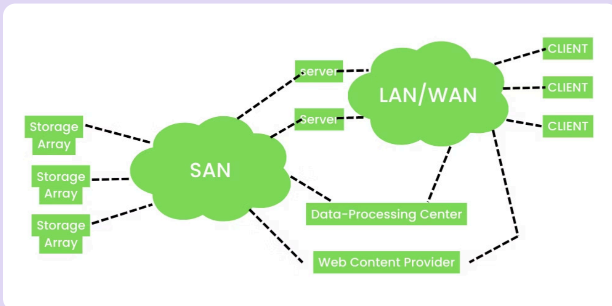


NAS has its own processor and memory so it can perform many tasks simultaneously without slowing down other devices on the network; this makes it an ideal solution for businesses who need high availability but don't have enough CPU power available at their desktops or laptops

3

StorageArea Network (SAN)

Storage area networks (SAN) are a network of storage devices that can be used to store and retrieve data from a shared central repository. A SAN is usually used for large data storage and retrieval, which requires high availability, scalability, reliability, and performance. The most common type of SAN uses fiber channel adapters to connect the host with its disk arrays.



A SAN can be implemented in two ways: as an external or internal device. An external SAN is used to connect storage devices to a network, while an internal SAN is integrated into the operating system of a computer. A disk is basically a device that communicates with the host to fetch data or store data in it with the help of I/O devices.

Disk Scheduling

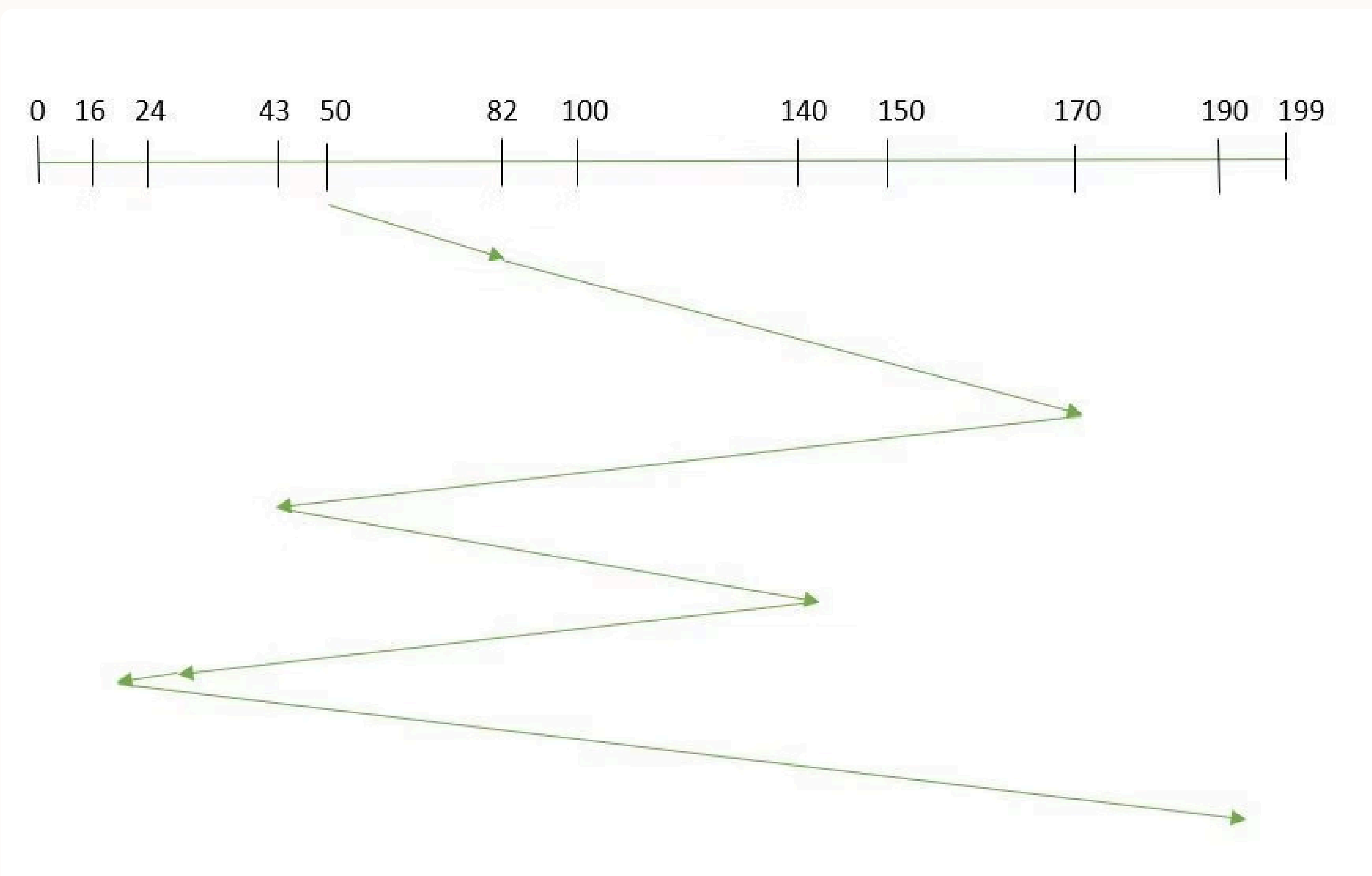
Different types of disk scheduling algorithms are as follows:

1. FCFS (First Come First Serve)
2. SSTF (Shortest Seek Time First)
3. SCAN (Elevator)
4. C-SCAN
5. LOOK
6. C-LOOK

FCFS scheduling algorithm:

This is the simplest form of disk scheduling algorithm. This services the request in the order they are received. This algorithm is fair but do not provide fastest service. It takes no special care to minimize the overall seek time.

Eg:- consider a disk queue with request for i/o to blocks on cylinders. 98, 183, 37, 122, 14, 124, 65, 67



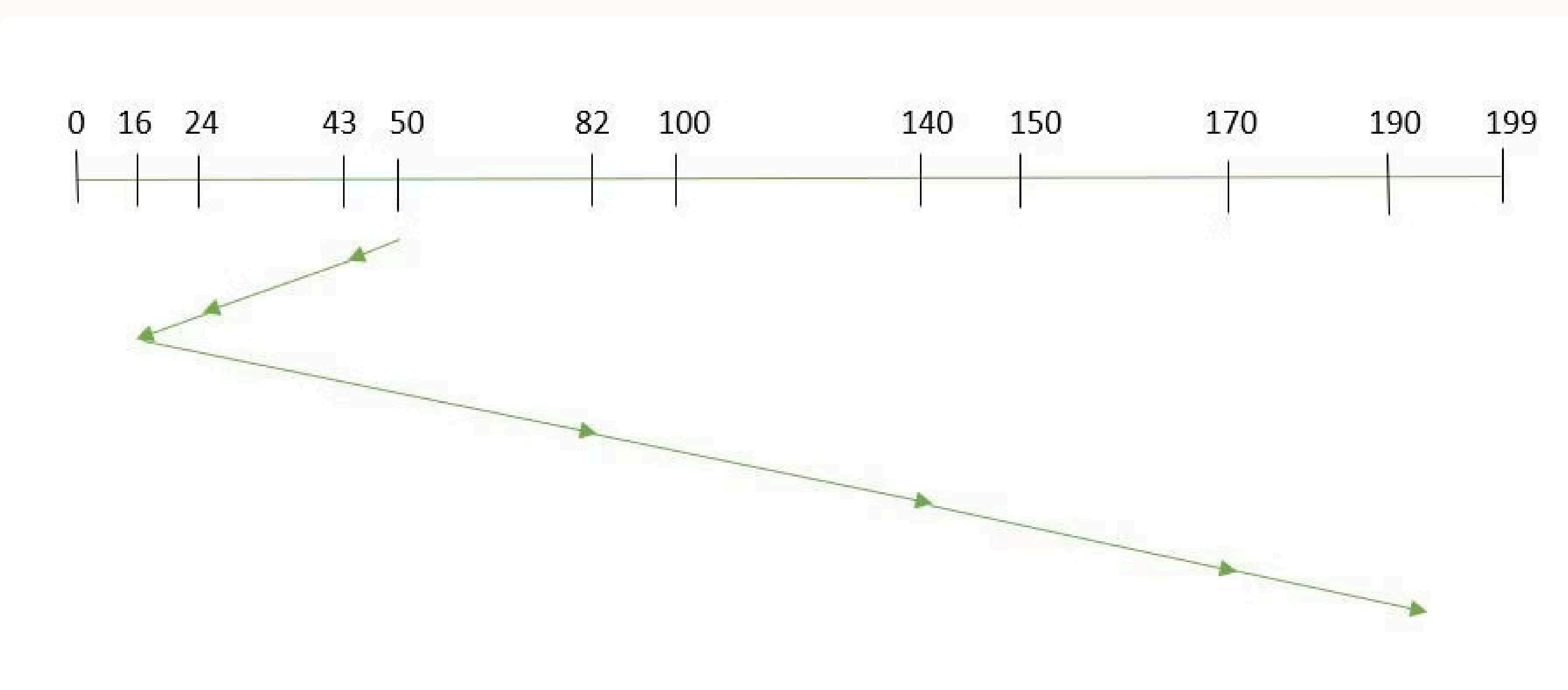
Suppose the order of request is- (82,170,43,140,24,16,190) and current position of Read/Write head is: 50

> *So, total overhead movement (total distance covered by the disk arm) = (82-50)+(170-82)+(170-43)+(140-43)+(140-24)+(24-16)+(190-16) =642*

SSTF

SSTF (Shortest Seek Time First) is a disk scheduling algorithm that selects the request closest to the current disk arm position, minimizing seek time. It improves performance compared to FCFS by reducing average response time and increasing system throughput.

- Selects the request with minimum seek time
- Executes requests closest to the disk head first
- Reduces average seek time and improves efficiency
- May cause starvation of distant requests



Suppose the order of request is- (82,170,43,140,24,16,190) and current position of Read/Write head is: 50

> *total overhead movement (total distance covered by the disk arm) = (50-43)+(43-24)+(24-16)+(82-16)+(140-82)+(170-140)+(190-170) =208*